

Paper 2 - Set 01 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	B	Differentiation
2	B	Indices and surds
3	C	Quadratic functions
4	C	The logic of arguments
5	E	Pythagoras, mensuration and 3D
6	D	The logic of arguments
7	A	Mathematical proof
8	F	Statistics
9	E	Mathematical proof
10	A	Identifying errors in proofs
11	F	Probability
12	F	Functions and transformations
13	A	Identifying errors in proofs
14	A	Identifying errors in proofs
15	H	Binomial expansion
16	G	The logic of arguments
17	E	Trigonometric functions and equations
18	E	Exponentials and logarithms
19	A	Mathematical proof
20	F	Chains of reasoning

Worked solutions

Question 1 DIF-17 · Differentiation · spec MM6.3

The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

Answer: B

$$P = (-k, 0). y' = 2x + k = -2k + k = -k = 6, \text{ so } k = -6.$$

Question 2 IND-04 · Indices and surds · spec MM1.2

Express $\frac{5}{3 + 2\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational.

Answer: B

$$\text{Multiply top and bottom by } 3 - 2\sqrt{5}: \text{ denominator } 9 - 20 = -11, \text{ numerator } 15 - 10\sqrt{5}. \text{ So the value is } \frac{15 - 10\sqrt{5}}{-11} = \frac{-15 + 10\sqrt{5}}{11}.$$

Question 3 QUA-24 · Quadratic functions · spec MM1.3

Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

Answer: C

$$(x - 4)(x + 2) < 0 \text{ gives } -2 < x < 4; \text{ the integers are } -1, 0, 1, 2, 3: \text{ five of them.}$$

Question 4 LGC-07 · The logic of arguments · spec Arg2

Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?

Answer: C

A square has four equal sides ($P \Rightarrow Q$, sufficient), but a rhombus that is not a square shows Q does not imply P (not necessary).

Question 5 MEN-11 · Pythagoras, mensuration and 3D · spec M5.16

Which of the following statements is/are true?

- I Doubling the radius of a circle doubles its circumference.
- II Doubling the radius of a sphere multiplies its volume by 8.
- III Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

Answer: E

$$C = 2\pi r \text{ is linear in } r \text{ (I). } V \propto r^3 \text{ (II). } \pi(2r)^2 \cdot \frac{h}{2} = 2\pi r^2 h: \text{ doubled, not unchanged (III false).}$$

Question 6 LGC-21 · The logic of arguments · spec Arg3

Consider the statement S : '**for all real x , $x^2 - 6x + 10 > 0$.' Which of the following is true?

Answer: D

Completing the square shows the expression is at least 1 for every x .

Question 7 X2-27 · Mathematical proof · spec Prf1

Consider the claim: ****if n is a positive integer then $n^2 + n + 1$ is odd.**** Which of the following is a valid proof?

Answer: A

A is a direct general argument. B only checks cases; C is irrelevant; D is correct for odd n but says nothing about even n ; E does not reach a contradiction.

Question 8 STA-05 · Statistics · spec M6.4

Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?

- I Students who revised more tended to score higher.
- II Revising more causes a higher mark.
- III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.

Answer: F

Correlation describes a tendency (I) and supports interpolation within the data range (III), but does not by itself establish causation (II).

Question 9 PRF-04 · Mathematical proof · spec Prf1

A student wants to prove that ****the square of any odd integer leaves remainder 1 when divided by 8****. Which of the following is the most appropriate first step?

Answer: E

A direct proof: $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$, and $k(k + 1)$ is even, so the square is $8m + 1$. Checking cases is not a proof; assuming the claim is circular.

Question 10 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

- I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.
- II $(x + 1)^2 \geq 0$ for all real x .
- III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .
- IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Question 11 PRO-03 · Probability · spec M7.7

$P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I A and B are independent.
- II A and B are mutually exclusive.
- III $P(A \text{ or } B) = 0.8$

Answer: F

$P(A)P(B) = 0.3 = P(A \text{ and } B)$, so independent (I). Not mutually exclusive since they can both happen (II false).
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$ (III).

Question 12 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$ for $x \geq 3$; true. $f(g(x)) = \sqrt{x^2} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Question 13 ERR-11 · Identifying errors in proofs · spec Err1

Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

- I An antiderivative of x^{-2} is $-x^{-1}$.
- II So the integral equals $[-\frac{1}{x}]_{-1}^1 = -1 - 1 = -2$.
- III Hence the value is -2 .

Which of the following is correct?

Answer: A

The integrand is positive everywhere it is defined, so a negative answer is a warning sign; the function is not defined on the whole interval and the integral does not exist in the ordinary sense.

Question 14 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

- I $0.1 > 0.01$.
- II Taking logarithms to base 10 of both sides: $-1 > -2$.
- III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Question 15 BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
- II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
- III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1+1)^n$ expanded. All true.

Question 16 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I For all x and all y , if $x < y$ then $x^2 < y^2$.
- II For all x there exists y such that $x < y$ and $x^2 < y^2$.
- III There exist x and y with $x < y$ and $x^2 > y^2$.

Answer: G

I fails for $x = -2, y = 1$. II: take $y = |x| + 1$. III: $x = -2, y = 1$ works.

Question 17 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of $y = \cos x$ is/are true?

- I It is symmetric about the y -axis.
- II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
- III It has period π .

Answer: E

$\cos(-x) = \cos x$ so I is true. $\sin(x + \frac{\pi}{2}) = \cos x$, so II is true. The period of $\cos$ is 2π , so III is false.

Question 18 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim '**for all positive x and y ,

$\log_{10}(x+y) \leq \log_{10} x + \log_{10} y$ '**?

Answer: E

$x = y = 1$: $\log 2 > 0 = \log 1 + \log 1$, so the inequality fails. For the others $x + y \leq xy$ and the claim holds (log is increasing).

Question 19 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: '**if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.'** Which of the following proves it?

Answer: A

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

Question 20 RSN-08 · Chains of reasoning · spec Prf5

a , b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

Answer: F

I: $a + b < c + c$. III: $\frac{a+c}{2} < \frac{c+c}{2} = c$. II fails when b is negative: $a = -3$, $b = -2$, $c = -1$ gives $ab = 6 > bc = 2$.